

A Characterization of State Spaces of Orthomodular Lattices

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It is shown that every rational polytope is affinely equivalent to the set of all states of a finite orthomodular lattice, and that every compact convex subset of a locally convex topological vector space is affinely homeomorphic to the set of all states of an orthomodular lattice.

INTRODUCTION

In axiomatic quantum mechanics one framework often used is that of an orthomodular lattice together with a set of states on the lattice. One then adds "physically reasonable" axioms to ensure that the orthomodular lattice is isomorphic to the lattice of closed subspaces of Hilbert space and/or that the set of states is affinely equivalent to the set of positive trace class operators of unit trace [7, 8]. Another approach begins with an abstract state space and focuses on its affine and topological structure [1, 3, 4].

In this paper we answer a question connecting the two approaches: which sets (up to affine and topological equivalence) can occur as the set of all states on an orthomodular lattice? A related question has been answered by Davies [2] who observed that any compact convex subset of a locally convex (topological vector) space can be represented as an order determining subset of the states on an orthomodular lattice. Greechie [5] has shown that the empty set is representable: There exists a (finite) orthomodular lattice which admits no states. It is not difficult to see that the set of all states of a finite orthomodular lattice is a rational polytope (a polytope whose vertices have all rational coordinates); we will show that every rational polytope is affinely equivalent to the set of states of a finite orthomodular lattice. Furthermore, we will show that any compact convex subset of a locally convex space is affinely homeomor-

phic to the set of all states of an orthomodular lattice; the converse is evident. It will also be shown that every compact convex subset of a locally convex space is affinely homeomorphic to the set of all countably additive states on a suitable σ -orthomodular lattice; the converse is false.

DEFINITIONS AND BASIC CONCEPTS

Let L be a lattice with greatest element 1 and least element 0. Recall that a map $\prime: L \rightarrow L$ is an *orthocomplementation* if for all x and y in L (i) $(x')' = x$; (ii) $x \vee x' = 1$, and (iii) $x \leq y$ implies $y' \leq x'$. If L is a lattice and $\prime$ is an orthocomplementation on L , we say $(L, \prime)$ is an *orthomodular lattice* if $x \leq y$ implies $y = x \vee (y \wedge x')$ for all x and y in L .

Let L be an orthomodular lattice. We say L is finite if the underlying set is finite; L is a σ -orthomodular lattice if every countable subset of L has a least upper bound in L . If x and y are elements of L then x is *orthogonal* to y if $x \leq y'$. A *state* on L is a map $s: L \rightarrow I$ (where I is the unit interval in the reals) which satisfies $s(1) = 1$ and $s(x \vee y) = s(x) + s(y)$ whenever x is orthogonal to y . A set S of states on L is *order-determining* if x and y in L and $s(x) \leq s(y)$ for s in S implies $x \leq y$. A set S of states is *strongly order determining* if x and y in L and

$$\{s \in S \mid s(x) = 1\} \subseteq \{s \in S \mid s(y) = 1\}$$

implies $x \leq y$. The set of all states on L will be denoted $S(L)$. A state s on a σ -orthomodular lattice L is *countably additive* if $s(\bigvee x_i) = \sum s(x_i)$ for every sequence $\{x_i\}$ of pairwise orthogonal elements in L .

For us a locally convex space will be a locally convex Hausdorff topological vector space over the reals. If A and B are affinely homeomorphic subsets of locally convex spaces we will write $A \approx B$. If V is a locally convex space then V' will denote the set of continuous linear functionals from V into the reals R . If X is a topological space and A any set then X^A will denote the set of maps from A to X with the product topology.

Recall that a polytope is a bounded subset of R^n which is the intersection of a finite number of halfspaces (for us halfspace will mean closed halfspace). A *halfspace* H in R^n is *rational* if H can be written in the form

$$H = \left\{ (x_1, \dots, x_n) \in R^n \mid \sum a_i x_i \leq c \right\}$$

with $a_1, \dots, a_n, c$ rational numbers. A *polytope* is *rational* if it is the intersection of a finite number of rational halfspaces; this happens if and only if its vertices have all rational coordinates. (If a polytope is rational

then each extreme point (being the intersection of the bounding hyperplanes containing it [9, Theorem 2.4.7]) must have rational coordinates. If the vertices of a polytope have all rational coordinates, one can explicitly write the polytope as an intersection of rational halfspaces [9, pp. 58–9]).

We will now describe a useful procedure for constructing orthomodular lattices, due to Greechie [5]. Let X be a nonempty set (whose members will be called *points*) and $\mathcal{F}$ a collection of finite, nonempty subsets of X (members of $\mathcal{F}$ are *frames*). An n -tuple ($n > 2$) $(F_1, \dots, F_n)$ of distinct frames of $\mathcal{F}$ is said to be a loop of length n if: (i) $F_i \cap F_j \neq \emptyset$ precisely when $|i - j| \in \{0, 1, n - 1\}$ and (ii) $F_i \cap F_j \cap F_k = \emptyset$ for all distinct i, j, k . We now define a G -space to be a pair $(X, \mathcal{F})$ (as described above) satisfying

- (i) Every frame has at least three members;
- (ii) if F_1 and F_2 are frames and $F_1 \neq F_2$ then $F_1 \cap F_2$ contains at most one point; and
- (iii) $(X, \mathcal{F})$ contains no loop of length less than five.

A convenient way of displaying a G -space is to represent points as points in the plane and each frame F as a line segment (or a smooth curve if necessary) containing the points in F (see Fig. 1).

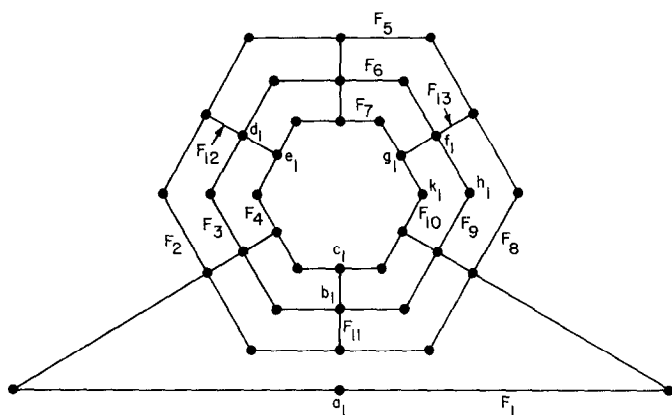


FIG. 1. The G -space $(X_1, \mathcal{F}_1)$.

If $(X, \mathcal{F})$ is a G -space there is a natural way to associate an orthomodular lattice L (this procedure is described in the proof of Lemma 1). Via this procedure certain mappings in I^X (called weights) correspond to the states on L . Let X be any nonempty set and $\mathcal{F}$ a collection of finite nonempty subsets of X ; a map $w: X \rightarrow I$ is said to be a *weight* on $(X, \mathcal{F})$

if $\sum_{x \in F} w(x) = 1$ for every frame F in $\mathcal{F}$. The set of all weights on $(X, \mathcal{F})$ is denoted $W(X, \mathcal{F})$. The correspondence of $W(X, \mathcal{F})$ and $S(L)$ is given in the following lemma; $W(X, \mathcal{F})$ and $S(L)$ are given the affine and topological structure inherited from R^X and R^L , respectively. Most of Lemma 1 and its proof can be found in [5, 6].

LEMMA 1. *If $(X, \mathcal{F})$ is a G -space then there exists a σ -orthomodular lattice L such that $S(L)$ is affinely homeomorphic to $W(X, \mathcal{F})$. L can be chosen so that every state on L is countably additive. If X is finite then L may be chosen to be finite.*

Proof. We first consider the case where $\mathcal{F}$ covers X . Let $\mathcal{O}$ be the collection of all subsets of frames $F \in \mathcal{F}$. Define a relation $\perp$ on X by $x_1 \perp x_2$ if $x_1 \neq x_2$ and $\{x_1, x_2\} \subseteq F$ for some frame F . If $A \subseteq X$, then define $A^\perp = \{x \in X \mid \text{if } a \in A, \text{ then } x \perp a\}$. Define $L = \{(A^\perp)^\perp \mid A \in \mathcal{O}\}$, and partially order L by inclusion. Define $': L \rightarrow L$ by $[(A^\perp)^\perp]' = [(A^\perp)^\perp]^\perp = A^\perp$. (The properties of a G -space guarantee that $A^\perp \in L$). Then $(L, \leq, ')$ is an orthomodular lattice, as was shown by Greechie [5], Theorem 3 (our notation is that of [6]). Furthermore, every pairwise orthogonal subset of L is finite (this follows from [5], Theorem 4) so that L is a σ -orthomodular lattice and every state on L is countably additive.

If s is a state on L , define $w_s: X \rightarrow I$ by $w_s(x) = s(\{(x)^\perp\}^\perp)$. Then w_s is a weight on $(X, \mathcal{F})$, and the map $s \mapsto w_s$ is an affine equivalence from $S(L)$ onto $W(X, \mathcal{F})$ [6], Theorem 1.6. The map is clearly continuous, and since $S(L)$ is a closed (therefore compact) subset of I^L , and $W(X, \mathcal{F})$ is Hausdorff, then $S(L)$ is affinely homeomorphic to $W(X, \mathcal{F})$.

We now consider the case where $\mathcal{F}$ does not cover X . Let Y be the set of points in X not contained in any frame. Let Y' be a copy of Y , disjoint from X , and let $y \mapsto y'$ be a 1-1 correspondence from Y onto Y' . Now define $X' = X \cup Y'$, and $\mathcal{F}' = \mathcal{F} \cup \{\{y, y'\} \mid y \in Y\}$. If $w \in W(X, \mathcal{F})$ define $w': X' \rightarrow I$ by $w'|_X = w$, and $w'(y') = 1 - w(y)$ for all y in Y . Then it is not difficult to verify that $w \mapsto w'$ is an affine homeomorphism from $W(X, \mathcal{F})$ onto $W(X', \mathcal{F}')$.

Since some frames of $(X', \mathcal{F}')$ have only two points, then $(X', \mathcal{F}')$ is not a G -space. Nevertheless, the procedure outlined at the start of this proof still yields a σ -orthomodular lattice L (the horizontal sum of the σ -orthomodular lattice which corresponds to the G -space $(X \setminus Y, \mathcal{F})$ and $|Y|$ copies of the 4-element Boolean lattice). Finally, the map $s \mapsto w_s$ described above is again an affine homeomorphism from $S(L)$ onto $W(X', \mathcal{F}')$, and therefore $S(L)$ and $W(X, \mathcal{F})$ are affinely homeomorphic.

CHARACTERIZATION OF STATE SPACES

THEOREM. *If C is a compact convex subset of a locally convex space, then C is affinely homeomorphic to the set of all states on a σ -orthomodular lattice L ; if C is a rational polytope L can be chosen to be finite. Conversely, if L is an orthomodular lattice, then the state space $S(L)$ of L is a compact convex subset of I^L ; if L is finite then $S(L)$ is a rational polytope.*

Remarks. Mr. Eric Gerelle (personal communication) is responsible for the observation that the state space of a finite orthomodular lattice is a rational polytope.

So that the relevance of the following lemmas will be more evident, we will now outline the proof of the theorem. The compact convex set C can be assumed to be a subspace of I^X for a suitable set X , and C is equal to the intersection of some collection $\{H_\alpha\}$ of half-spaces in R^X . Each H_α can be written in the form

$$H_\alpha = \left\{ w \in R^X \mid \sum a_i w(x_i) \leq c \right\}$$

for a suitable choice of real numbers $c, a_1, \dots, a_n$, and $\{x_1, \dots, x_n\} \subseteq X$. Note that $I^X = W(X, \emptyset)$; Lemma 3 then guarantees the existence of a G -space $(X_\alpha, \mathcal{F}_\alpha)$ (for each α) with $W(X_\alpha, \mathcal{F}_\alpha) \approx H_\alpha \cap I^X$; these G -spaces can be chosen so that $(\bigcup X_\alpha, \bigcup \mathcal{F}_\alpha)$ is a G -space with $W(\bigcup X_\alpha, \bigcup \mathcal{F}_\alpha) \approx (\bigcap H_\alpha) \cap I^X = C$. The theorem then follows from Lemma 1.

We will now state and prove Lemma 2 (used in proving Lemma 3) and Lemma 3. (Note that in conjunction with Lemma 1, Lemma 2 yields an orthomodular lattice with exactly one state, generalizing the similar result for orthomodular posets due to Greechie and Miller [6]). We will then give the complete proof of the theorem.

LEMMA 2. *There exists a finite G -space $(X, \mathcal{F})$ which admits exactly one weight w ; this G -space can be chosen so that $w(x) = 1$ for at least one x in X .*

Proof. Let $(X_1, \mathcal{F}_1)$ be the G -space portrayed in Fig. 1. (The author wishes to thank the referee for suggesting this space, a space which is less complicated than the author's original one). Observe that the frames $\{F_i, i = 1, \dots, 13\}$ form a disjoint cover of X_1 ; it follows that if w is a weight on $(X_1, \mathcal{F}_1)$ then

$$\sum \{w(x) \mid x \in X_1\} = 13.$$

On the other hand, the remaining twelve frames in $\mathcal{F}_1$ form a disjoint

cover of $X_1 - \{a_1\}$; it follows that for every weight w on $(X_1, \mathcal{F}_1)$ we have $w(a_1) = 1$. Observe also that there is exactly one weight w on $(X_1, \mathcal{F}_1)$ such that $w(b_1) = w(c_1) = w(d_1) = w(e_1) = w(f_1) = w(g_1) = w(h_1) = w(k_1) = 0$; the values of this weight are shown in Fig. 2.

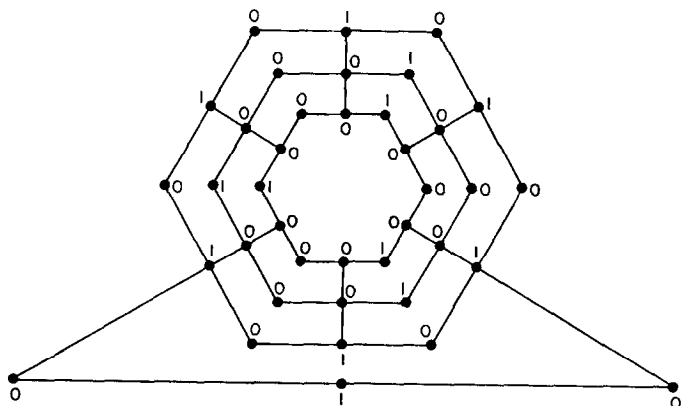


FIG. 2. The weight w on $(X_1, \mathcal{F}_1)$ with $w(b_1) = w(c_1) = w(d_1) = w(e_1) = w(f_1) = w(g_1) = w(h_1) = w(k_1) = 0$.

Now let $(X_2, \mathcal{F}_2), \dots, (X_9, \mathcal{F}_9)$ be disjoint copies of $(X_1, \mathcal{F}_1)$; label the points corresponding to $a_1, b_1, c_1, d_1, e_1, f_1, g_1, h_1, k_1$ as $a_i, b_i, \dots, k_i$ for $i = 2, \dots, 9$. Let $Z = \{z_1, \dots, z_{72}\}$ be a set of 72 elements disjoint from $\bigcup X_i$ (unless otherwise specified indices will run from $i = 1$ to $i = 9$). Define $X = (\bigcup X_i) \cup Z$ and

$$\begin{aligned} \mathcal{F} = & (\bigcup \mathcal{F}_i) \cup \{\{a_1, z_j, b_j\} \mid j = 2, \dots, 9\} \\ & \cup \{\{a_2, z_{j+9}, c_j\} \mid j \in \{1, 2, \dots, 9\} \setminus \{2\}\} \\ & \cup \{\{a_3, z_{j+18}, d_j\} \mid j \in \{1, 2, \dots, 9\} \setminus \{3\}\} \\ & \dots \\ & \cup \{\{a_8, z_{j+63}, k_j\} \mid j \in \{1, 2, \dots, 9\} \setminus \{8\}\} \\ & \cup \{\{a_9, z_1, b_1\}, \{a_9, z_{11}, c_2\}, \{a_9, z_{21}, d_3\}, \dots, \{a_9, z_{71}, k_8\}\}. \end{aligned}$$

Suppose that w is a weight on $(X, \mathcal{F})$. Then w restricted to any $(X_i, \mathcal{F}_i)$ is also a weight, so $w(a_i) = 1$ for $i = 1, \dots, 9$. Furthermore, since $\{a_1, z_2, b_2\}$ is a frame then $w(z_2) = w(b_2) = 0$; in a similar manner it follows that w maps all of Z to zero, and $w(b_i) = w(c_i) = w(d_i) = w(e_i) = w(f_i) = w(g_i) = w(h_i) = w(k_i) = 0$ for $i = 1, \dots, 9$. From the remark made previously it follows that the restriction of w to $(X_1, \mathcal{F}_1)$ must agree with the values portrayed in Fig. 2 and similarly for each $(X_i, \mathcal{F}_i)$, $i = 2, \dots, 9$. Conversely, define $w: X \rightarrow I$ so that w restricted to $(X_1, \mathcal{F}_1)$ has the

values in Fig. 2, w restricted to X_i (for $i = 2, \dots, 9$) has the corresponding values, and so that w maps Z to zero. Then w is a weight on $(X, \mathcal{F})$.

Finally, we observe that $(\bigcup X_i, \bigcup \mathcal{F}_i)$ is clearly a G -space, and the additional points and frames added to get $(X, \mathcal{F})$ yield no loops of length less than five so $(X, \mathcal{F})$ is a G -space with exactly one weight.

LEMMA 3. Assume that $(X, \mathcal{F})$ is a G -space, $x_1, \dots, x_n$ are members of X , and $c, a_1, \dots, a_n$ are real numbers. Then there exists a G -space $(X', \mathcal{F}')$ with $X \subseteq X'$ and $\mathcal{F} \subseteq \mathcal{F}'$ such that:

(i) Every weight w on $(X', \mathcal{F}')$ restricts to a weight on $(X, \mathcal{F})$, and satisfies

$$a_1 w(x_1) + \dots + a_n w(x_n) \leq c; \quad (1)$$

and

(ii) every weight w on $(X, \mathcal{F})$ satisfying (1) extends uniquely to a weight on $(X', \mathcal{F}')$.

Furthermore, if $c, a_1, \dots, a_n$ are rational numbers, then X' and $\mathcal{F}'$ can be chosen so that the sets $X' \setminus X$ and $\mathcal{F}' \setminus \mathcal{F}$ are finite.

Proof. Suppose that the lemma holds for a particular choice of constants $c, a_1, \dots, a_n$. We will then say that one can force $\sum a_i w(x_i) \leq c$; we shall use similar language if the inequality is replaced with equality.

Now let x be any point in X . We first observe that one can force $w(x) = 0$. To see this let $(X_L, \mathcal{F}_L)$ denote a G -space possessing exactly one weight and with, for example, $w(a_1) = 1$ for some point a_1 in X_L . Now define $X' = X \cup X_L \cup \{p\}$ and $\mathcal{F}' = \mathcal{F} \cup \mathcal{F}_L \cup \{(a_1, p, x)\}$; we assume X and X_L are disjoint and $\{p\}$ is a singleton set disjoint from $X \cup X_L$. It can be easily checked that this construction forces $w(x) = 0$; we represent this construction by the diagram in Fig. 3(a).

Now let x and y be two distinct points in a G -space $(X, \mathcal{F})$. By adjoining the points and frames in Fig. 3(b), one forces $w(x) = w(y)$. [In Fig. 3(b), only x and y are in the original G -space $(X, \mathcal{F})$. Note that we cannot get by with a shorter chain from x to y , since we might then

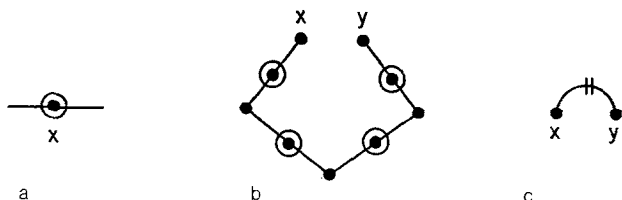


FIG. 3. Constructions for Lemma 3.

have a loop of length less than five in $(X', \mathcal{F}')$.] Figure 3(c) gives an abbreviated representation of the construction forcing $w(x) = w(y)$. In Fig. 4(a) a construction forcing $w(x) = (m/n)w(y)$ (for m and n positive integers) is portrayed. We have assumed in Fig. 4(a) that $m \geq 2$, $n \geq 2$, and $n > m$; clearly this involves no loss of generality.

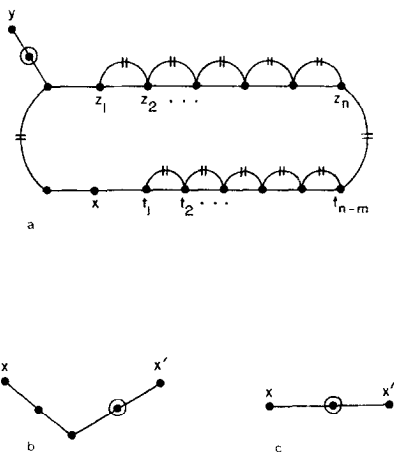


FIG. 4. More constructions for Lemma 3.

Now let $(X, \mathcal{F})$ be any G -space and x any point in X . The construction in Fig. 4(b) gives a G -space $(X', \mathcal{F}')$ with $X' \supseteq X$, $\mathcal{F}' \supseteq \mathcal{F}$, and a point x' in $X' \setminus X$ such that:

(i) Every weight w on $(X, \mathcal{F})$ extends to a weight w' on $(X', \mathcal{F}')$; and

(ii) for every number $c \geq w(x)$ there exists a unique extension w' of w such that $w'(x') = c$ (assuming $c \leq 1$).

We will refer to this construction as “adjoining a point x' and forcing $w(x) \leq w(x')$.”

Now we can force $w(x) \leq (m/n)w(y)$. To accomplish this, adjoin a point x' , force $w(x) \leq w(x')$, and then force $w(x') = (m/n)w(y)$.

Note that in all constructions so far $X' \setminus X$ and $\mathcal{F}' \setminus \mathcal{F}$ are finite. Now we relax this constraint, and give a construction forcing $w(x) = \lambda w(y)$ for any positive real number λ . Choose a sequence $\{p_n\}$ of rational numbers approaching λ monotonically from above. For $n = 1, 2, \dots$ let $(X_n, \mathcal{F}_n)$ be the G -space described above which forces $w(x) \leq p_n w(y)$; choose these spaces so that the sets $X_n \setminus X$ are mutually disjoint and similarly for the sets $\mathcal{F}_n \setminus \mathcal{F}$. Now let $X' = \bigcup X_i$ and $\mathcal{F}' = \bigcup \mathcal{F}_i$. Then

$(X', \mathcal{F}')$ is a G -space which forces $w(x) \leq \lambda w(y)$. By also forcing $w(y) \leq (1/\lambda) w(x)$ we can then force $w(x) = \lambda w(y)$.

We need one more forcing technique. Let $(X, \mathcal{F})$ be a G -space and x any point in X . Figure 4(c) portrays the construction of a G -space $(X', \mathcal{F}')$ with $X' \supseteq X$, $\mathcal{F}' \supseteq \mathcal{F}$, and a point x' in $X' \setminus X$ such that:

(i) Every weight w on $(X, \mathcal{F})$ extends to a unique weight w' on $(X', \mathcal{F}')$;

(ii) this unique extension satisfies $w'(x) + w'(x') = 1$.

We describe this construction as “adjoining x' and forcing $w(x) + w(x') = 1$.”

Finally, let $x_1, \dots, x_n$ and $c, a_1, \dots, a_n$ be as in the statement of the lemma. Define sets of integers P and N by $i \in P$ if $a_i \geq 0$, $i \in N$ if $a_i < 0$. For each i in N adjoin a point x'_i and force $w(x_i) + w(x'_i) = 1$. Then forcing $\sum a_i w(x_i) \leq c$ is equivalent to forcing

$$\sum_{i \in P} a_i w(x_i) + \sum_{i \in N} |a_i| w(x'_i) \leq c - \sum_{i \in N} a_i;$$

thus, without loss of generality we may assume $a_i \geq 0$ for $i = 1, \dots, n$. Now if $c < 0$ there are no weights w on $(X, \mathcal{F})$ satisfying $\sum a_i w(x_i) \leq c$; therefore adjoin a point y and force $w(x_1) + w(y) = 1$ and $w(x_1) = w(y) = 0$; this “forces” $\sum a_i w(x_i) \leq c$. Thus we can assume $c \geq 0$. If $\sum a_i = 0$, then $\sum a_i x_i \leq c$ holds for all weights on $(X, \mathcal{F})$ and we can choose $X' = X$, $\mathcal{F}' = \mathcal{F}$; we therefore assume $\sum a_i > 0$ and without loss of generality we assume $a_i > 0$ for $i = 1, \dots, n$. Now if $c = 0$ $\sum a_i w(x_i) \leq c$ precisely if $w(x_i) = 0$ for all i ; the appropriate construction was discussed previously. Finally, assume $c > 0$; dividing by c we may assume $c = 1$. Let $\{u_i \mid 1 \leq i \leq n+2\}$ be a set disjoint from X , let $(X_1, \mathcal{F}_1) = (X \cup \{u_1, \dots, u_{n+2}\}, \mathcal{F} \cup \{\{u_1, \dots, u_{n+2}\}\})$. Now force $w(u_{n+2}) = 0$, $w(x_i) = (1/a_i) w(u_i)$ for $1 \leq i \leq n$. Then if $(X', \mathcal{F}')$ is the resulting G -space, each weight w on $(X', \mathcal{F}')$ satisfies $\sum a_i w(x_i) = \sum_{i=1}^n w(u_i) \leq 1 = c$ and $(X', \mathcal{F}')$ is a G -space which forces $\sum a_i w(x_i) \leq c$.

Proof of the Theorem. Let L be an orthomodular lattice. It follows from the definition of a state that $S(L)$ is a convex subset of I^L and is closed (therefore compact) in I^L . If L is finite then $S(L)$ is a bounded subset of R^L and is determined by a finite set of linear equations and inequalities with rational coefficients ($0 \leq s(x) \leq 1$ for all x in L , $s(1) = 1$, and $s(x \vee y) = s(x) + s(y)$ if x is orthogonal to y) so $S(L)$ is a rational polytope.

Conversely, suppose C is a compact convex subset of a locally convex

space. We will assume C is a subspace of $I^X \subseteq R^X$ for some set X . (This involves no loss of generality since the evaluation map is an affine homeomorphism of C into $I^X \subseteq R^X$ where X is the set of continuous affine maps from C into I .) Note that if C is a polytope we can assume X is finite. Since C is closed and convex, then C is the intersection of half-spaces, say $C = \bigcap H_\alpha$ where

$$H_\alpha = \{w \in R^X \mid f_\alpha(w) \leq a_\alpha\}$$

with $f_\alpha \in (R^X)'$.

Let $F(X)$ denote the free vector space on X . Then $F(X)$ and R^X are put in duality by the bilinear form $\langle \sum \lambda_i x_i, f \rangle = \sum \lambda_i f(x_i)$. The $w(R^X, F(X))$ topology on R^X makes the dual of R^X precisely $F(X)$; since this topology is just the product topology it follows that $(R^X)' = F(X)$. Thus every halfspace H in R^X is of the form

$$H = \left\{ w \in R^X \mid \sum a_i w(x_i) \leq c \right\}$$

for some choice of real numbers $a_1, \dots, a_n, c$. If C is a rational polytope, then C is the intersection of a finite number of halfspaces which can be described with rational coefficients $a_1, \dots, a_n, c$.

Note that $(X, \emptyset)$ is a G -space, whose set of weights is I^X . By Lemma 3, for each halfspace H_α there exists a G -space $(X_\alpha, \mathcal{F}_\alpha)$ with $X_\alpha \supseteq X$ such that:

(i) Every weight on $(X, \emptyset)$ in $H_\alpha \cap I^X$ extends uniquely to a weight on $(X_\alpha, \mathcal{F}_\alpha)$;

(ii) every weight on $(X_\alpha, \mathcal{F}_\alpha)$ restricts to a weight on $(X, \emptyset)$ which is in $H_\alpha \cap I^X$.

We can choose the spaces $(X_\alpha, \mathcal{F}_\alpha)$ so that the sets $X_\alpha \setminus X$ are mutually disjoint, and similarly the sets $\mathcal{F}_\alpha$. Now define $X' = \bigcup X_\alpha$ and $\mathcal{F}' = \bigcup \mathcal{F}_\alpha$. Then:

(i) Every weight on $(X, \emptyset)$ in $C = \bigcap (H_\alpha \cap I^X)$ extends uniquely to a weight on $(X', \mathcal{F}')$;

(ii) every weight on $(X', \mathcal{F}')$ restricts to a weight on $(X, \emptyset)$ in C .

The restriction map from $W(X', \mathcal{F}')$ onto C is clearly an affine equivalence. Since $X \subseteq X'$, the map is also continuous and therefore [since $W(X', \mathcal{F}')$ is compact and C is Hausdorff] the restriction map is a homeomorphism.

Finally, if C is a rational polytope then X , each $X_\alpha \setminus X$, and each $\mathcal{F}_\alpha$

can be chosen to be finite, so X' and $\mathcal{F}'$ are finite. The theorem now follows from Lemma 1.

FINAL COMMENTS

From the proof just completed and Lemma 1 it follows that if C is any compact convex subset of a locally convex space then there exists a σ -orthomodular lattice L such that the set of all countably additive states on L is affinely equivalent to C . However, the converse is not true, as may be seen from the following example. Let L be the Borel subsets of R modulo sets of Lebesgue measure zero. Then L is a Boolean σ -algebra (and therefore certainly a σ -orthomodular lattice). Let $S_c(L)$ be the set of countably additive states on L ; then $S_c(L)$ has no extreme points and therefore by the Krein–Milman theorem $S_c(L)$ is not affinely equivalent to any compact convex subset of a locally convex space.

In closing we mention that the following question is of interest and remains open: How can one characterize the state space $S(L)$ of an orthomodular lattice L if L is assumed to admit an order-determining (or, strongly order-determining) set of states? We conjecture that there do exist compact convex subsets of locally convex spaces not representable in either manner.

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REFERENCES

1. E. M. ALFSEN, "Compact Convex Sets and Boundary Integrals," Springer-Verlag, Heidelberg, 1971.
2. E. B. DAVIES, Example related to the foundations of quantum theory, *J. Math. Phys.* **13** (1972), 39–41.
3. E. B. DAVIES AND J. T. LEWIS, An operational approach to quantum probability, *Commun. Math. Phys.* **17** (1970), 239–260.
4. C. M. EDWARDS, The theory of pure operations, *Commun. Math. Phys.* **24** (1972), 260–288.
5. R. J. GREECHIE, Orthomodular lattices admitting no states, *J. Combinatorial Theory* **10** (1971), 119–132.
6. R. J. GREECHIE AND F. R. MILLER, On structures related to states on an empirical logic I. Weights on finite spaces, Technical Report 14, Department of Mathematics, Kansas State University, Manhattan, Kansas, April 1970.

7. J. M. JAUCH AND C. PIRON, On the structure of quantal proposition systems, *Helv. Phys. Acta.* **42** (1969), 842–848.
8. G. LUDWIG, An improved formulation of some theorems and axioms in the axiomatic foundation of the Hilbert space structure of quantum mechanics, *Commun. Math. Phys.* **26** (1972), 78–86.
9. J. STOER AND C. WITZGALL, “Convexity and Optimization in Finite Dimensions I,” Springer-Verlag, Heidelberg, 1970.